

Magnetostatică

$$F = \frac{\mu \cdot I_1 \cdot I_2 \cdot l}{2\pi d}, \quad \mu = \mu_0 \cdot \mu_r, \quad \mu_0 = 4\pi \cdot 10^{-7} \frac{\text{H}}{\text{m}}$$

$$F = B \cdot I \cdot l \cdot \sin \alpha$$

$$\vec{B} = \mu \cdot \vec{H}$$

$$B = \frac{\mu I}{2\pi r} \rightarrow \text{la distanța „r” de un conductor liniar}$$

$$B = \frac{\mu I}{2r_0} \rightarrow \text{în centrul spirii}$$

$$B = \frac{\mu N I}{L} \rightarrow \text{pentru solenoid}$$

Forța Lorentz: $\vec{F} = q \cdot \vec{v} \times \vec{B}$
 $\vec{F} = q \vec{E} + q \vec{v} \times \vec{B}$ (în câmp electromagnetic)

$$\Phi_m = B \cdot S \cdot \cos(\vec{B}, \vec{S}) \quad (\text{fluxul magnetic})$$

$$I = \frac{Q}{t}$$

$$i = \frac{e}{R}, \quad i = \frac{\Delta Q}{\Delta t}$$

$$e = - \frac{\Delta \Phi}{\Delta t} \rightarrow \text{Legea inducției electromagnetice}$$

$$\text{div } \vec{B} = 0 \rightarrow \text{legea lui Gauss în magnetostatică}$$

$$\text{div } \vec{E} = \frac{\rho}{\epsilon_0} \rightarrow \text{legea lui Gauss în electrostatică}$$

* Fizica solidului

$$I = I_0 \left(e^{\frac{eV}{kT}} - 1 \right), \quad e = 1,6 \cdot 10^{-19} \text{ C}$$

$$k = 1,38 \cdot 10^{-23} \text{ J/K (const. Boltzmann)}$$

caracteristica tipului de semiconductor

T - temperatură absolută

e - sarcina electronului

$$R_T = R_\infty \cdot e^{\frac{b}{T}} \quad \left. \vphantom{R_T = R_\infty \cdot e^{\frac{b}{T}}} \right\} \text{Termistor}$$

$$R_T = R_0 \cdot e^{\frac{b}{T} - \frac{b}{T_0}}$$

Dinamica punctului material

$$\vec{F} = \frac{d\vec{p}}{dt}, \quad \vec{F} = 0 \Rightarrow \frac{d\vec{p}}{dt} = 0 \Rightarrow \vec{p} = \text{constant} \quad (\text{Teorema impulsului})$$

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\vec{M} = \vec{r} \times \vec{F}$$

$$\vec{M} = \frac{d\vec{L}}{dt} \quad (\text{Teorema momentului cinetic})$$

$$\vec{M} = 0 \Rightarrow \vec{L} = \text{constant}$$

$$\delta W = \vec{F} \cdot d\vec{r} = F dr \cos \theta \quad (\text{Lucrul mecanic elementar})$$

$$\delta W = dE_c \quad (\text{legea de variație a energiei cinetice})$$

$$\vec{F} = -\nabla U = -\left(\frac{\partial U}{\partial x} \vec{i} + \frac{\partial U}{\partial y} \vec{j} + \frac{\partial U}{\partial z} \vec{k}\right) \quad (\text{Forța conservativă})$$

$$\vec{F}(x) = -\frac{dU}{dx} \rightarrow \text{mişcare liniară}$$

$$\delta W = -\nabla U \cdot d\vec{r}, \quad d\vec{r} = \vec{i} dx + \vec{j} dy + \vec{k} dz, \quad \delta W = -dU$$

$$v = \frac{ds}{dt}$$

$$a = \frac{dv}{dt}$$

$$d = v \cdot t \quad (\text{mişcare uniformă})$$

$$\vec{F} = m \cdot \vec{a}$$

$$\left. \begin{aligned} x &= x_0 + v_0 t + \frac{1}{2} a t^2 \\ v &= v_0 + at \end{aligned} \right\} \rightarrow \text{mişcare uniform accelerată (a-est.)}$$

$$x = x_0 + vt \rightarrow \text{mişcare rectilinie uniformă (v-est.)}$$

$$\left. \begin{aligned} x &= x_0 + v_0 t + \frac{1}{2} a t^2 \\ v &= v_0 + at \end{aligned} \right\} \rightarrow \text{mişcare uniform încetinită (a-est.)}$$

Electrostatică

• Legea lui Coulomb:

$$\vec{F} = k \cdot \frac{q_1 q_2}{r_{12}^2} \cdot \vec{r}_{12}, \quad k = \frac{1}{4\pi\epsilon}, \quad \epsilon = \epsilon_0 \cdot \epsilon_r, \quad \epsilon_0 = \frac{1}{4\pi \cdot 9 \cdot 10^9} \Rightarrow$$

$$\vec{F} = k \cdot \frac{q_1 q_2}{r_{12}^2}$$

$$F = \frac{q_1 q_2}{4\pi \epsilon r^2}$$

$$\Rightarrow \epsilon_0 = 8,85 \cdot 10^{-12} \frac{C^2}{N \cdot m^2}$$

• Intensitatea câmpului electric

$$\vec{E} = \frac{\vec{F}_0}{q_0} = \frac{1}{4\pi \epsilon_0} \cdot \frac{Q}{r^2} \cdot \vec{r}, \quad \vec{E} = \sum_{i=1}^n \vec{E}_i$$

$$E = \frac{Q}{4\pi \epsilon_0 r^2}$$

• Potentialul electric

$$\varphi(\vec{r}) = - \int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{r} = \frac{Q}{4\pi \epsilon r}$$

$$\vec{E} = -\text{grad } \varphi$$

• Legea lui Gauss

$$\iint_S \vec{E} \cdot d\vec{s} = \frac{q_{\text{int}}}{\epsilon}$$

$$\text{div } \vec{E} = \frac{\rho}{\epsilon}$$

• Fluxul electric:

$$\Phi = \iint_S \vec{E} \cdot d\vec{s}$$

• Fluxul electric elementar

$$d\Phi_e = \vec{E} \cdot d\vec{s}_m = E \cdot s \cdot \cos \alpha$$

$$\vec{D} = \epsilon \cdot \vec{E}$$

$$D = \frac{Q}{2\pi r_0}$$

$$\begin{cases} E_x = \frac{d\varphi}{dx} \\ E_y = \frac{d\varphi}{dy} \\ E_z = \frac{d\varphi}{dz} \end{cases}$$

Oscilații mecanice. 1. Mișcarea oscilatorie ideală, armonică

$$1. \omega_0^2 = \frac{k}{m}, \quad \omega_0 = \sqrt{\frac{k}{m}}, \quad k = m \omega_0^2$$

$$x(t) = A \cdot \sin(\omega_0 t + \varphi_0) = \text{legea de mișcare}$$

$$v(t) = \frac{dx}{dt}, \quad a(t) = \frac{dv}{dt}$$

$$E_c = \frac{1}{2} \cdot m \cdot v^2$$

$$E_p = \frac{1}{2} k \cdot x^2$$

$$E = E_c + E_p = \frac{1}{2} k A^2 = \frac{1}{2} m \omega_0^2 A^2 \quad (\text{energia totală})$$

$$\omega_0 = \frac{2\pi}{T_0}, \quad \omega_0 = 2\pi \nu$$

$$T_0 = 2\pi \sqrt{\frac{l}{g}} \rightarrow \text{în cazul pendulului gravitațional}$$

$$T_0 = 2\pi \sqrt{\frac{m}{k}} \rightarrow \text{în cazul pendulului elastic}$$

2. Mișcarea oscilatorie amortizată

$$\omega = \sqrt{\omega_0^2 - \rho^2} \Rightarrow x(t) = \underbrace{A_0 \cdot e^{-\rho t}}_{A(t)} \sin(\omega \cdot t + \varphi)$$

$$\Delta = \ln \frac{A(t)}{A(t+T)} = \ln e^{\rho T} = \rho \cdot T, \quad e = 2, 71$$

$$E = \frac{1}{2} k A^2 = \frac{1}{2} k A_0^2 \cdot e^{-2\rho t}$$

$$\frac{E(t)}{E(t+T)} = 2, 718 = e, \quad T - \text{timpul de relaxare}$$

$$2\rho \cdot T = 1$$

$$T = \frac{1}{2\rho} \Rightarrow T = \frac{m}{g}, \quad A = A_0 \cdot e^{-\rho t}$$

3. Oscilații forțate

$$F_p = F_0 \cdot \sin(\omega_p t) \rightarrow \text{forța perturbatoare}$$

$$y(t) = y_0(t) + y_p(t)$$

$$y_0(t) = A_0 e^{-\rho t} \sin(\omega t + \varphi)$$

$$y_p(t) = \frac{F_0}{f} \sin(\omega_p t - \varphi)$$

$$\frac{F_0 T}{(\omega_0^2 - \omega_p^2)^2 + (2\rho \omega_p)^2}, \quad \tan \varphi = \frac{2\rho \omega_p}{\omega_0^2 - \omega_p^2}$$

$$f = \frac{F_0}{m}, \quad 2\rho = \frac{g}{m}$$

$$\Delta E_c = \mathcal{L} = F \cdot x$$

• Rezonanță

$$A_{\text{max}} = A_{\text{max}} = \frac{f}{2\beta\sqrt{\omega_0^2 - \beta^2}}$$

$$\omega_p = \omega_{\text{rez}} = \sqrt{\omega_0^2 - 2\beta^2}$$

1. Compunerea oscilațiilor ($\omega_1 = \omega_2$)

$$y(t) = y_1(t) + y_2(t)$$

$$y_1(t) = A_1 \sin(\omega t + \varphi_{01})$$

$$y_2(t) = A_2 \sin(\omega t + \varphi_{02})$$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_{02} - \varphi_{01})}$$

$$\tan \varphi_0 = \frac{y}{x} = \frac{y_1 + y_2}{x_1 + x_2} = \frac{A_1 \sin(\varphi_{01}) + A_2 \sin(\varphi_{02})}{A_1 \cos(\varphi_{01}) + A_2 \cos(\varphi_{02})} \Rightarrow$$

$$\Rightarrow y(t) = A \sin(\omega t + \varphi_0)$$

$$A = A_1 + A_2, \text{ dacă } \Delta\varphi = 0 \Rightarrow A_{\text{max}} \text{ (în fază)}$$

$$A = |A_1 - A_2|, \text{ dacă } \Delta\varphi = \pi \Rightarrow A_{\text{min}} \text{ (în opoziție de fază)}$$

$$A = \sqrt{A_1^2 + A_2^2}, \text{ dacă } \Delta\varphi = \frac{\pi}{2} \text{ (în cuadratură de fază)}$$

2. Compunerea oscilațiilor armonice paralele de frecvență diferită

$$\left. \begin{aligned} y_1(t) &= A_1 \sin(\omega_1 t + \varphi_1) \\ y_2(t) &= A_2 \sin(\omega_2 t + \varphi_2) \end{aligned} \right\} \Rightarrow y(t) = A \sin(\omega t + \varphi)$$

$$y(t) = y_1(t) + y_2(t), \omega_1 = \omega + \Delta\omega, \omega_2 = \omega - \Delta\omega,$$

$$A^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos(2\Delta\omega t + \varphi_1 - \varphi_2); \tan \varphi = \frac{A_1 \sin(\omega t + \varphi_1) - A_2 \sin(\omega t + \varphi_2)}{A_1 \cos(\omega t + \varphi_1) + A_2 \cos(\omega t + \varphi_2)}$$

• Fenomenul de bătăi

$$A_1 = A_2 = A_0 \text{ și } \varphi_1 = \varphi_2 = 0 \Rightarrow A = 2A_0 \cos(\Delta\omega t)$$

$$T_b = \frac{2\pi}{\Delta\omega} = \frac{2\pi}{\frac{\omega_1 - \omega_2}{2}} \quad y = 2A_0 \cos\left(\frac{\omega_1 - \omega_2}{2} t\right) \sin\left(\frac{\omega_1 + \omega_2}{2} t\right)$$

3. Compunerea oscilațiilor perpendiculare

$$x(t) = A_1 \sin(\omega t + \varphi_1)$$

$$y(t) = A_2 \sin(\omega t + \varphi_2)$$

$$\left(\frac{x}{A_1}\right)^2 + \left(\frac{y}{A_2}\right)^2 - 2\frac{x}{A_1} \frac{y}{A_2} \cos(\varphi_2 - \varphi_1) = \sin^2(\varphi_2 - \varphi_1)$$

↳ ecuația generalizată a elipsei